

Unanimity over two utilities

Model — human; produced by literature.

0.1 Package

- **Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \succcurlyeq q$ and $q \succcurlyeq r$ then $p \succcurlyeq r$.
- **Independence.** For all $p, q, r \in \Delta(X)$ and $\lambda \in (0,1]$: $p \succcurlyeq q$ if and only if $\lambda p + (1-\lambda)r \succcurlyeq \lambda q + (1-\lambda)r$.
- **Closed-graph continuity.** The set $\{(p, q) \in \Delta(X) \times \Delta(X) : p \succcurlyeq q\}$ is closed in the product topology.
- **Expected multi-utility representation.** There is a set U of functions $u : X \rightarrow \mathbb{R}$ such that for all $p, q \in \Delta(X)$: $p \succcurlyeq q$ if and only if $EU_u(p) \geq EU_u(q)$ for every $u \in U$.
- $\neg$ **Completeness.** For all $p, q \in \Delta(X)$: $p \succcurlyeq q$ or $q \succcurlyeq p$.

0.2 Construction

Unanimity over two utilities. With $X = \{a, b, c\}$ let $p \succcurlyeq q$ iff $p(a) \geq q(a)$ and $p(b) \geq q(b)$ ($U = \{u_1, u_2\}$ with $u_1 = 1_a$, $u_2 = 1_b$). This is an expected multi-utility representation, hence transitive, independent and closed-graph continuous; but δ_a and δ_b are incomparable.

0.3 Sources

- Aumann (1962), Utility theory without the completeness axiom, *Econometrica* 30
- Dubra, Maccheroni & Ok (2004), *JET* 115

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